

A **variable in JavaScript** is a simple container used to store data so that it can be used later in a program. It has a name (called identifier) and a value stored inside it. Variables are created using **let**, **const**, or **var**.

When JavaScript runs the program, it creates space in memory and links the variable name to that value. Variables can store different types of data like numbers, text, or objects, and they help in performing calculations and building logic in programs.

In simple words...

JavaScript me **variables** ko simple language me samjhe to yeh ek container ya dabba (box) hota hai jisme hum data store karte hain, taaki baad me use kar sake.

variables programming ka basic building block hai—agar **variables** nahi honge to na data store hoga, na calculations, na logic properly kaam karega.

Think of **variables in JavaScript** like **English pronouns** such as *he*, *she*, *it*, *they*. Instead of repeating a name again and again, we use a pronoun to refer to it easily. Similarly, in programming, instead of writing the actual value again and again, we store it in a variable and use its name.

👉 **Analogy:**

If you say:

“Ved is a student. Ved studies in college. Ved likes coding.”

It sounds repetitive.

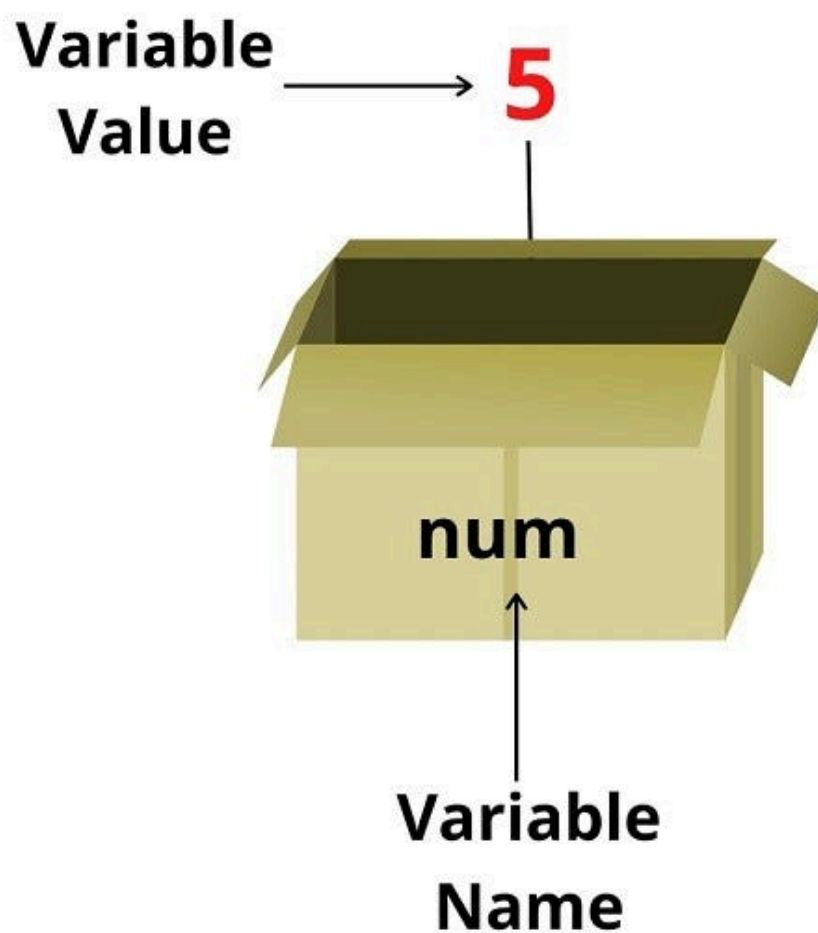
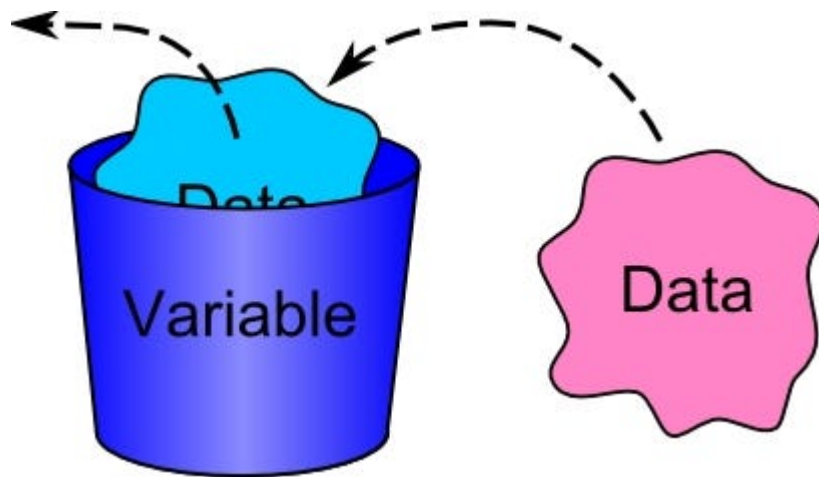
So we replace it with pronouns:

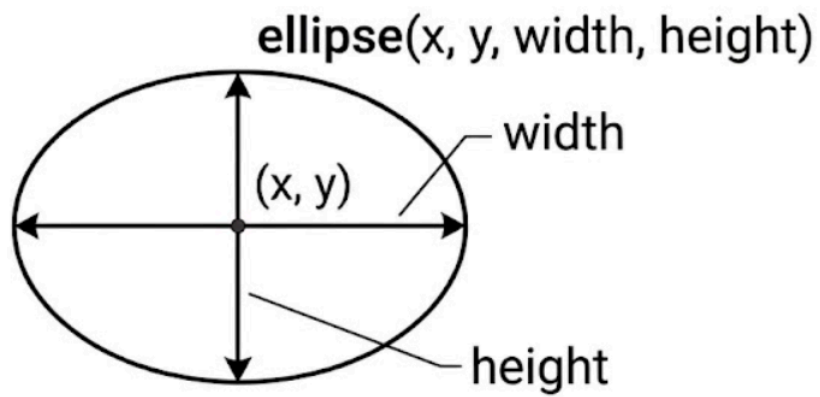
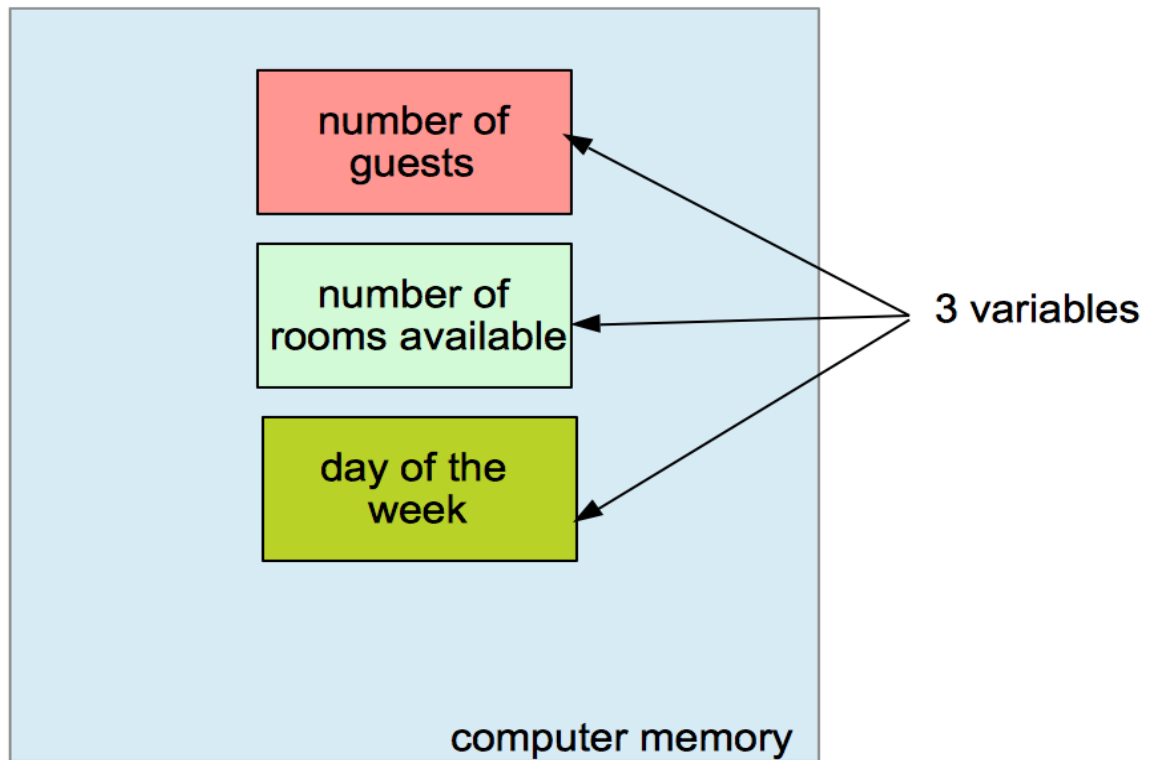
“Ved is a student. **He** studies in college. **He** likes coding.”

Here, “**he**” = **reference to Ved**

👉 Same way, **variable** = **reference to stored value**

INTRO TO JS: VARIABLES





Variable = Dabba of Chocolates

You have a **dabba (box)** full of chocolates.

- The **dabba ka naam** = variable name
- The **chocolates inside** = value (data)

Real Life 	Programming 
Dabba ka label	Variable name
Andar chocolates	Value
Dabba	Memory location

Example

👉 Maan lo:

```
var chocolates = 10;
```

🔍 Samjho:

- `chocolates` = dabba ka naam
- `10` = andar ke chocolates
- Memory me ek jagah reserve ho gayi

Change Value (Refill Dabba)

```
chocolates = 20;
```

👉 Ab:

- Same dabba
- Bas chocolates badh gaye 😊

Bohot Saare Dabbey = Multiple Variables

 **Imagine:**

Tumhare paas ek nahi, **bohot saare dabbey** hai:

- Ek me chocolates 
- Ek me biscuits 
- Ek me candies 

Har dabbe pe **alag naam** likha hai

 Programming Mapping	
Real Life 	Programming 
Chocolate dabba	<code>chocolates</code>
Biscuit dabba	<code>biscuits</code>
Candy dabba	<code>candies</code>

Example Code

```
var chocolates = 10;  
var biscuits = 5;  
var candies = 20;
```

 **Samjho:**

- 3 alag variables = 3 alag dabbe
- Har ek ke andar alag value

 **Value Change (Alag-Alag Control)**

```
chocolates = 15;  
biscuits = 8;
```

👉 Sirf selected dabba change hua

👉 Baaki same rehte hai 😎

Remember!!!

ellipse(x, y, width, height);

Maan lo tumhe 3 circle banane hai same position pe

✗ WITHOUT VARIABLE

```
ellipse(200, 150, 50, 50);  
ellipse(200, 150, 50, 50);  
ellipse(200, 150, 50, 50);
```

🔍 **Samjho:**

- Tumne **200, 150** baar baar likha
- 3 circles same jagah pe ban gaye

🚫 **Problem:**

👉 Agar sir bole:

“sab circles ko thoda right shift karo (300, 150)”

Toh tumhe:

```
ellipse(300, 150, 50, 50);  
ellipse(300, 150, 50, 50);  
ellipse(300, 150, 50, 50);
```

✗ Har line change karni padegi → time waste + mistake chance

✓ WITH VARIABLE

```
var x = 200;  
var y = 150;  
  
ellipse(x, y, 50, 50);  
ellipse(x, y, 50, 50);  
ellipse(x, y, 50, 50);
```

🔍 Samjho:

- **x** aur **y** me position store hai
- Tum repeat nahi kar rahe numbers

✨ Ab sir bole:

“position change karo (300, 150)”

👉 Bas yeh change karo:

```
x = 300;
```

✓ Aur automatically:

- teeno circles shift ho jayenge 😎
- sirf **1 jagah change**

⚡ CLEAR DIFFERENCE

👉 Without variable:

- “Har baar number likho” 😓

👉 With variable:

- “Ek naam use karo, sab control me” 😎

With Colouring Functions...

```
fill(255, 0, 0);    // Red
ellipse(150, 150, 100, 100);

fill(0, 255, 0);    // Green
ellipse(300, 150, 100, 100);

fill(0, 0, 255);    // Blue
ellipse(450, 150, 100, 100);
```

With variable use...

```
var r = 240;
var g = 100;
var b = 50;

fill(r, g, b);
ellipse(250, 150, 120, 120);
```

Colour change easily hoga....

```
r = 0;
g = 255;
b = 0;
```


Multiple Variable example...

```
var r1 = 255, g1 = 0, b1 = 0;  
var r2 = 0, g2 = 255, b2 = 0;  
var r3 = 0, g3 = 0, b3 = 255;  
  
fill(r1, g1, b1);  
ellipse(150, 150, 100, 100);  
  
fill(r2, g2, b2);  
ellipse(300, 150, 100, 100);  
  
fill(r3, g3, b3);  
ellipse(450, 150, 100, 100);
```

- `var r, g, b` = 3 color ke dabbe
- Value change karo → color change ho jata hai
- Same code reuse hota hai

💡 Simple line:

"RGB variables se hum color ko control kar sakte hai easily."